



On Diffusion Posterior Sampling

The Hidden Bias and Instability of Guided Diffusion

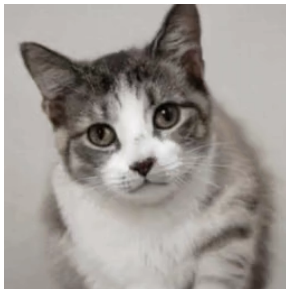
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Motivation

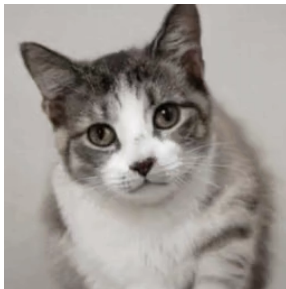
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We are all aware of image generation, but these advances have also been applied to protein synthesis, text generation, and many other fields.

Sohl-Dickstein et al. ICML '15; Ho, Jain, Abbeel NeurIPS '20; Song et al. ICLR '21; Rombach et al. CVPR '22.



Set up

We have access to data:

$$\{X_i\}_{i=1}^N \subset \mathbb{R}^K.$$

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How do we create a model that can generate new cat images?

We assume that the data are i.i.d. samples of some unknown underlying **target distribution**

$$X_i \sim \rho_* \in \mathcal{P}(\mathbb{R}^K).$$



Dynamic interpretation of sampling

We are interested in finding a dynamic process that converts a sample

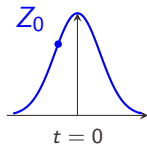
$$Z \sim \gamma = \mathcal{N}(0, I)$$

into a sample from the target distribution

$$X \sim \rho_*.$$

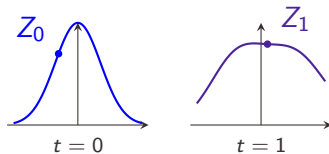


Time-Evolution of the distribution



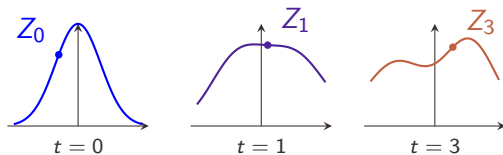


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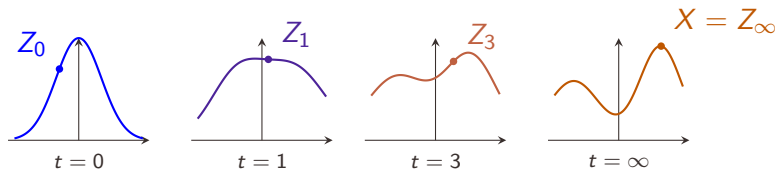


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The Sampling Problem

The problem is to find a vector field

$$V : [0, \infty) \times \mathbb{R}^K \rightarrow \mathbb{R}^K$$

such that Z_t evolves according to an SDE of the form

$$dZ_t = V(t, Z_t)dt + \sqrt{2}dB_t,$$



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and

$$\text{Law}(Z_\infty) \sim \rho_*.$$



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$$dX_t = -X_t dt + \sqrt{2} dB_t,$$

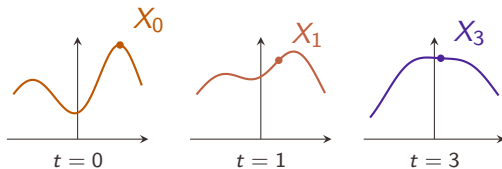


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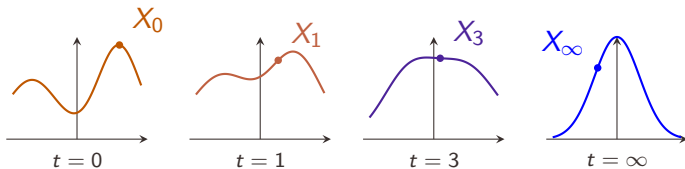


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Fokker-Planck equation

More precisely, the evolution of the law of $X_t \sim \vec{\rho}_t$ is given by the Fokker-Planck equation

$$\begin{cases} \partial_t \vec{\rho}_t = \nabla \cdot (\vec{\rho}_t x) + \Delta \vec{\rho}_t, \\ \vec{\rho}_0 = \rho_*. \end{cases}$$



Convergence Estimate

Exponential Convergence

$$\mathcal{H}(\vec{\rho}_t \mid \gamma) \leq Ce^{-t}.$$

for some constant C depending on the second moment of ρ_* , where $\mathcal{H}$ is the relative entropy.



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This estimate is independent of the regularity of ρ_* !



The backward process

We chose a time horizon $T > 0$, such that

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We consider the equation satisfied by the time-reversal of the law:

$$\overleftarrow{\rho}_t = \vec{\rho}_{T-t}.$$



The wrong backward process

WARNING

If we are not careful, using the standard Fokker-Planck equation

$$\begin{cases} \partial_t \vec{\rho}_t = \Delta \vec{\rho}_t + \nabla \cdot (\vec{\rho}_t x), \\ \vec{\rho}_0 = \rho_*, \end{cases}$$

we end up with the reverse heat equation

$$\begin{cases} \partial_t \overleftarrow{\rho}_t = -\Delta \overleftarrow{\rho}_t - \nabla \cdot (\overleftarrow{\rho}_t x), \\ \overleftarrow{\rho}_0 = \vec{\rho}_T. \end{cases}$$



The trick

We can re-write the Laplacian as a transport term:

$$\begin{aligned}\partial_t \vec{\rho} &= \Delta \vec{\rho} + \nabla \cdot (x \vec{\rho}) \\ &= -\Delta \vec{\rho} + 2\Delta \vec{\rho} + \nabla \cdot (x \vec{\rho}) \\ &= -\Delta \vec{\rho} + 2\nabla \cdot (\vec{\rho} \nabla \log \vec{\rho}) + \nabla \cdot (x \vec{\rho})\end{aligned}$$



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Reversing time, we get

$$\begin{cases} \partial_t \overleftarrow{\rho}_t = \Delta \overleftarrow{\rho}_t - 2\nabla \cdot (\overleftarrow{\rho}_t \nabla \log \vec{\rho}_{T-t}) - \nabla \cdot (x \overleftarrow{\rho}_t), \\ \overleftarrow{\rho}_0 = \vec{\rho}_T \sim \gamma. \end{cases}$$



Score Matching

Hence, we are left with finding a vector field V such that

$$\inf_{\Theta} \int_0^T \int_{\mathbb{R}^K} |V_{\Theta}(t, x) - \nabla \log \vec{\rho}_{T-t}(x)|^2 d\vec{\rho}_t dt \approx 0.$$



Why Score Matching Suffices

In fact, we have the quantitative bound

Error Estimate

Differentiating the relative entropy, we can show that

$$\begin{aligned}\mathcal{H}(\rho_* | \overleftarrow{\rho}_T^V) &\leq \mathcal{H}(\overrightarrow{\rho}_T | \gamma) \\ &\quad + \int_0^T \int_{\mathbb{R}^K} |V(t, x) - \nabla \log \overrightarrow{\rho}_{T-t}(x)|^2 d\overrightarrow{\rho}_t dt,\end{aligned}$$

where $\overleftarrow{\rho}_t^V$ is the solution of the Fokker-Planck equation

$$\begin{cases} \partial_t \overleftarrow{\rho}_t^V = \Delta \overleftarrow{\rho}_t^V + 2\nabla \cdot (V \overleftarrow{\rho}_t^V) + \nabla \cdot (x \overleftarrow{\rho}_t^V) \\ \overleftarrow{\rho}_0^V = \gamma. \end{cases}$$



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In our set up, we have access to samples $\{X_0^{(i)}\}_{i=1}^N$ from ρ_* .



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No functional representation of ρ_* is known a-priori!

However, we can produce samples from $\vec{\rho}_t$ cheaply. In particular, we have the explicit formula

$$X_t = e^{-t}X_0^i + \sqrt{1 - e^{-2t}}Z \sim \vec{\rho}_t$$

where

$$i \sim \text{uniform}\{1, \dots, N\}, \quad Z \sim \gamma.$$



Tweedie's formula

Using linearity of the equation, we can guess where we came from by looking at the current position and the time. More precisely, we have the formula

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We have a stochastic estimator of the score function $\nabla \log \vec{\rho}_t$ given by

$$\nabla \log \vec{\rho}_t(X_t) \approx \frac{1}{e^t - e^{-t}} (X_0 - e^t X_t).$$



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Sampling the initial condition X_0 from the data, the noise Z , and times, we can get an unbiased estimator of the score function.



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An option is always to do rejection sampling: generate unconditional samples and discard them until the feature appears.



Conditional Sampling

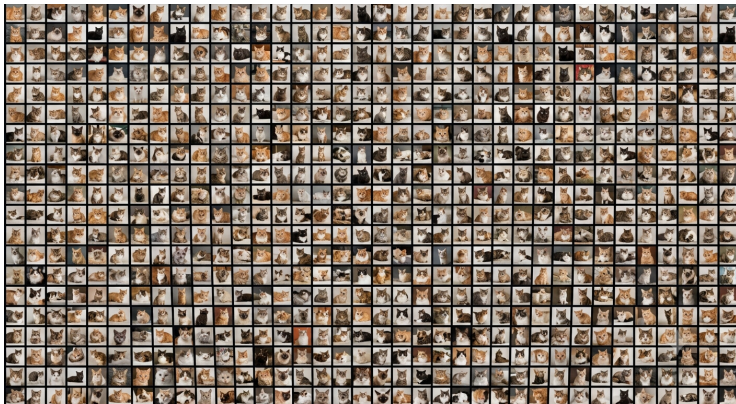
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The expected cost is $1/\text{Prob}(\text{feature})$, which can be very large for rare features.

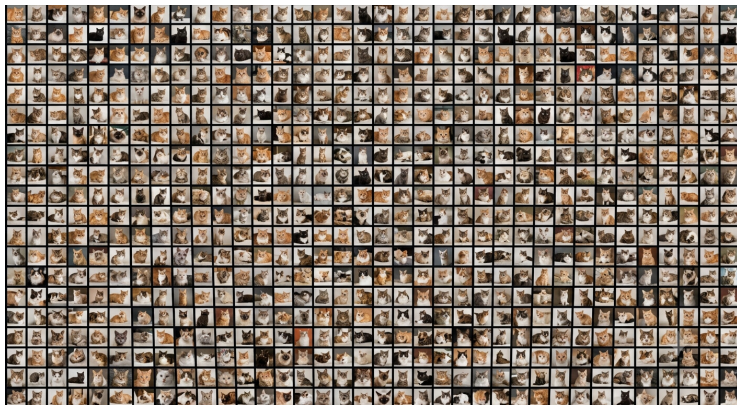


Rejection Sampling: Find a Calico!





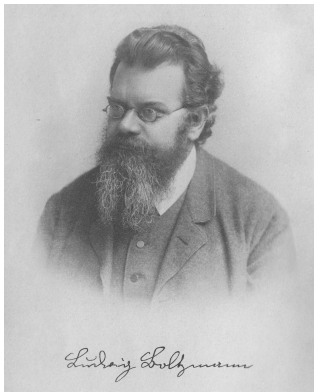
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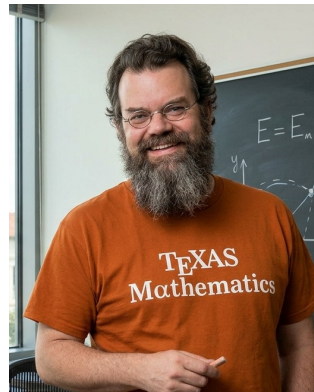
Each extra feature multiplies the cost: a calico *looking to the left* is twice as rare.



Conditional Sampling, in Practice



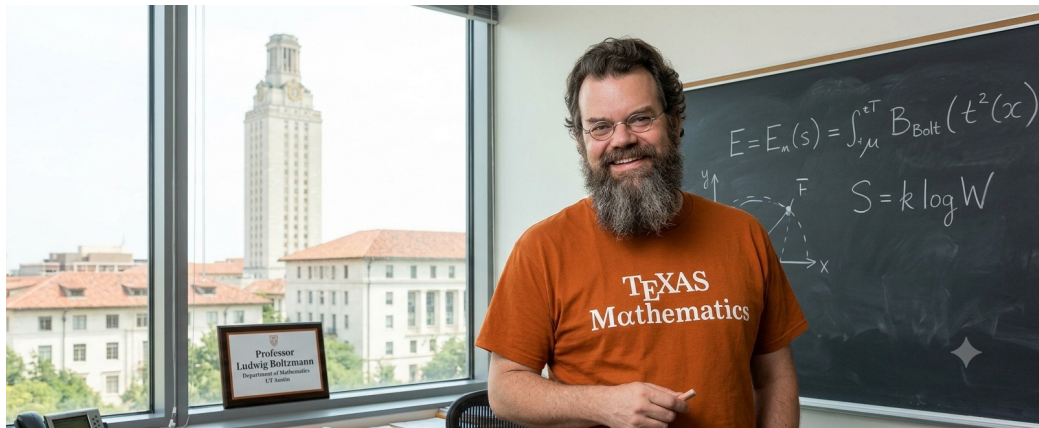
Prior sample: $x \sim \rho_*$ (Vienna, 1902)



Posterior sample: $x \sim \mu_y$, $y = \text{"UT faculty"}$

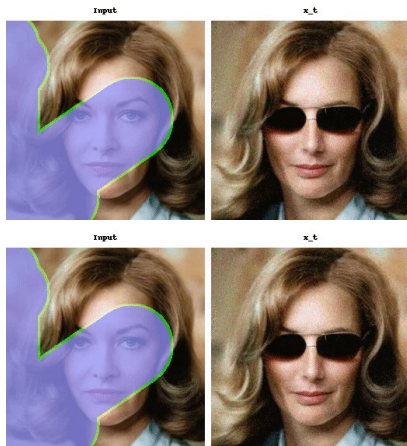


Welcome to UT, Prof. Boltzmann!





Inpainting





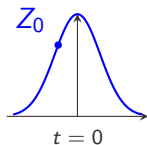
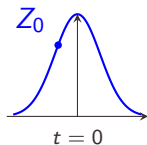
Super-resolution

Star Trek: The Next Generation style “Enhance!”.





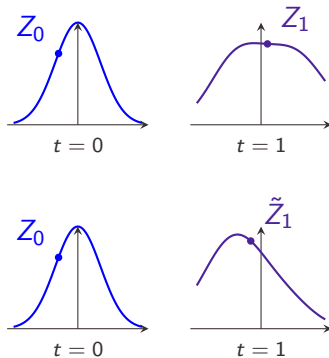
Conditional Dynamics: Mode Selection



$$\tilde{Z}_\infty \sim \rho_*(\cdot \mid \text{left well})$$



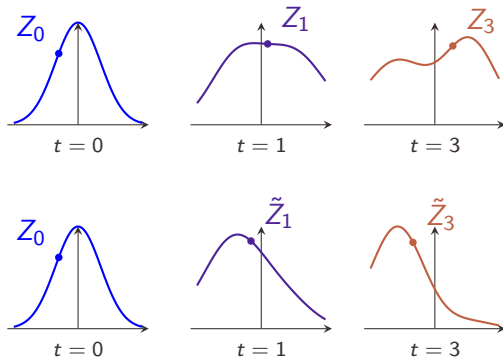
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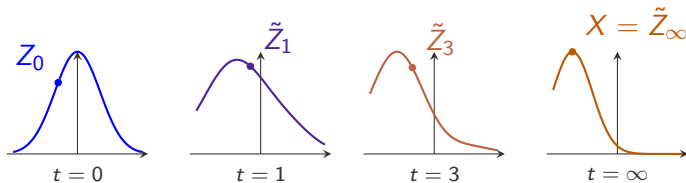
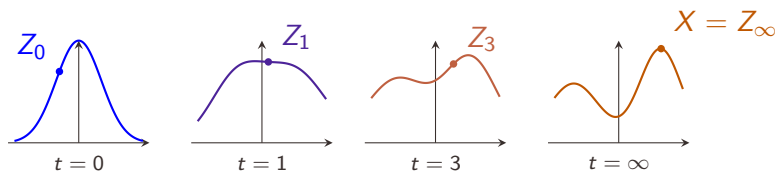
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For the inpainting problem, $\mathcal{A}(x) = Px$ where P is a projection on the observed pixels.
For example, dropping the second pixel:

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In super-resolution, $\mathcal{A}(x) = Ax$ is a projection on a lower resolution grid. For example, averaging neighboring pixels:

$$A = \frac{1}{4} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$



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Then, we can compute

$$\mu_y(x) = \frac{e^{-\frac{\|\mathcal{A}(x) - y\|_{\mathbb{R}^L}^2}{2\lambda}} \rho_*(x)}{Z}$$



In inverse problems, the set up is usually written as

$$y = \mathcal{A}(x) + \zeta,$$

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The relative entropy or the Gaussian noise can be replaced by other sensible choices.



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In theory, we could create a new dataset that is consistent with the measurement operator and learn the gradient of logarithm for O-U process with μ_y as initial condition.



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In theory, we could create a new dataset that is consistent with the measurement operator and learn the gradient of logarithm for O-U process with μ_y as initial condition.

The main point is to try avoid doing expensive computations again, even if we pay an acceptable bias. Can we recycle the score we have now precomputed to find a path from

$$\gamma \rightarrow \mu_y?$$



Surrogate Path

Problem: Find a path

$$\mu_t = g_t \vec{\rho}_t$$

(Surrogate Path)

such that

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That we can approximately invert.



First Try: OU

We can apply the same reasoning for the noising process:

$$\begin{cases} \partial_t \mu_t^{OU} = \Delta \mu_t^{OU} + \nabla \cdot (x \mu_t^{OU}) \\ \mu_0^{OU} = \mu_y. \end{cases}$$



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$$\mu_t^{OU}(x) = P(X_t = x|y) = \frac{\overbrace{P(y|X_t = x)}^{g_t(x)} P(X_t = x)}{P(y)}$$



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To invert the path, as before, we need to have an approximation of the vectorfield

$$\nabla \log \mu_t^{OU}(x) = \underbrace{\nabla \log P(y|X_t = x)}_{\nabla \log g_t(x)} + \underbrace{\nabla \log \vec{\rho}_t(x)}_{\text{Diffusion Oracle}}.$$



Exact solution

We know μ_t^{OU} and $\vec{\rho}_t$ solves O-U and

$$\mu_t^{OU} = \frac{g_t \vec{\rho}_t}{Z},$$

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$$\begin{cases} \partial_t g_t = \Delta g_t + x \nabla g_t + 2 \nabla \log \vec{\rho}_t g_t \\ g_0 = P(y|X_0 = x) = e^{-R_y(x)}. \end{cases}$$



Stochastic Characteristics

Moreover, we can use Stochastic Characteristics to solve

$$g_t(x) = \mathbb{E} [g_0(X_0) | X_t = x].$$



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Computational Complexity

This is too computationally heavy to be done in practice. Also, we need access to $\nabla \log g_t(x)$ which adds another layer of complication.



Taking approximations ML style

The original paper of Diffusion Posterior Sampling (DPS), just says that we can pass the expectation inside and hope for the best:

$$\mathbb{E}[g_0(X_0) | X_t = x] = g_t(x) \neq h_t(x) = g_0(\mathbb{E}[X_0 | X_t = x])$$



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The advantage is that the right hand side can be computed using the diffusion oracle

$$\mathbb{E}[X_0 | X_t = x] = e^t x + (e^t - e^{-t}) \nabla \log \vec{\rho}_t(x). \quad (\text{Tweedie's formula})$$



Ode to simplicity

We can consider

$$\mu_t^{OU} \neq \mu_t^{DPS} = \frac{\overbrace{e^{-R_y(\mathbb{E}[X_0|X_t=x])}}^{\text{Explicitly known}} \vec{\rho}_t}{Z_t}$$



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Using the properties of the conditional expectation

$$\lim_{t \rightarrow 0^+} \mu_t^{DPS} = \mu_y \quad \text{and} \quad \lim_{t \rightarrow \infty} \mu_t^{DPS} = \gamma.$$



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and we have direct access to compute

$$\nabla \log \mu_t^{DPS} = -\nabla(R_y(\mathbb{E}[X_0|X_t=x])) + \nabla \log \vec{\rho}_t$$

by knowing R_y and the diffusion oracle.



Property of the expected value

Let's understand the conditional expectation $\mathbb{E}[X_0|X_t = x]$ a bit better. We have the following property:

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So, for instance if $\rho_* = \delta_0$, then $\mathbb{E}[X_0|X_t = x] = 0$ for all $t \geq 0$ and $x \in \mathbb{R}^K$.

If $\rho_* = \frac{1}{2}(\delta_{-1} + \delta_1)$, then $\mathbb{E}[X_0|X_t = x] \in [-1, 1]$ for all $t \geq 0$ and $x \in \mathbb{R}^K$.



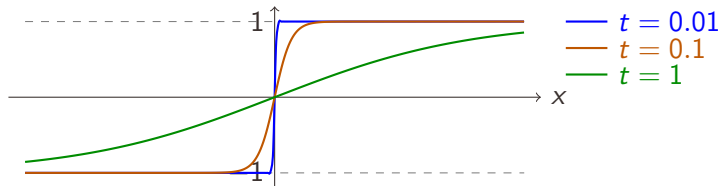
Binary prior: explicit posterior mean

Consider

$$dX_t = -X_t dt + \sqrt{2} dB_t, \quad X_0 \sim \frac{1}{2}(\delta_{-1} + \delta_1).$$

For fixed $t > 0$:

$$\mathbb{E}[X_0 \mid X_t = x] = \tanh\left(\frac{e^{-t}}{1 - e^{-2t}}x\right) = \tanh\left(\frac{x}{e^t - e^{-t}}\right).$$





DPS Algorithm

- ▶ Pick a time horizon T large enough.



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$$dY_t = \left(Y_t + 2\nabla \log \mu_{T-t}^{DPS}(Y_t) \right) dt + \sqrt{2}dB_t$$



DPS Algorithm

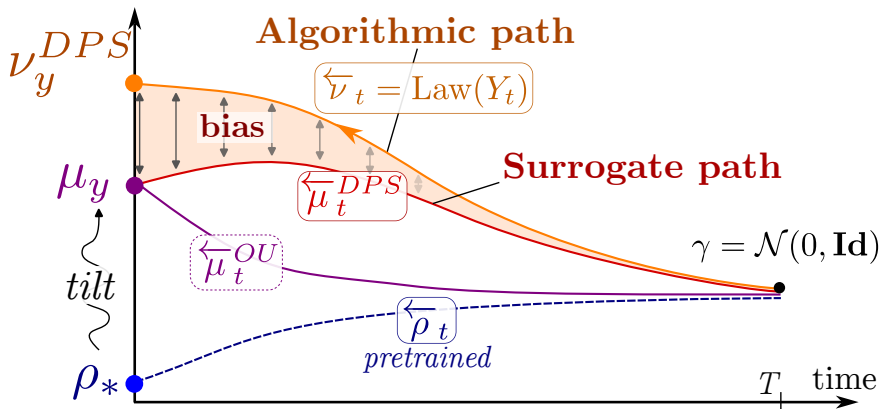
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- ▶ Use Y_T as a sample of ν_y^{DPS} .



Overview of the Construction





Where is the catch?

Biased sampler

DPS is not exact even for Gaussian measures!



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In a nutshell, μ_t^{DPS} does not satisfy O-U:

$$\begin{cases} \partial_t \mu_t^{DPS} = \Delta \mu_t^{DPS} + \nabla \cdot (x \mu_t^{DPS}) + (c_t - \mathbb{E}_{\mu_t^{DPS}} c_t) \mu_t^{DPS} \\ \mu_0^{DPS} = \mu_y \end{cases}$$

where

$$\begin{aligned} c_t(x) = & \text{trace}(\nabla \mathbb{E}[X_0 | X_t = x] D^2 R_y \nabla \mathbb{E}[X_0 | X_t = x]) \\ & - |\nabla R_y \nabla \mathbb{E}[X_0 | X_t = x]|^2 \end{aligned}$$



Seeing the Bias: a Gaussian Mixture

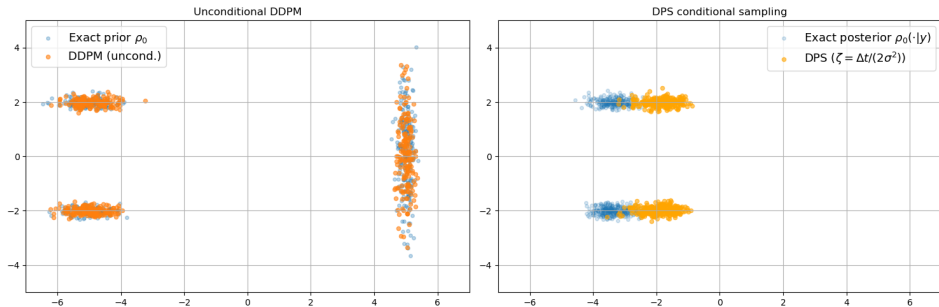


Figure: Gaussian-mixture prior. Left: unconditional DDPM samples match the prior. Right: DPS samples (orange) are systematically shifted with respect to the exact posterior (blue).



The reverse process for DPS

Given a fixed time horizon T , we can consider the equation satisfied by

$$\overleftarrow{\mu}_t^{DPS} = \mu_{T-t}^{DPS}.$$



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While $\overleftarrow{\nu}_t = \text{Law}(Y_t)$ satisfies

$$\begin{cases} \partial_t \overleftarrow{\nu}_t = \Delta \overleftarrow{\nu}_t - 2\nabla \cdot (\nabla \log \overrightarrow{\mu}_{T-t}^{DPS} \overleftarrow{\nu}_t) - \nabla \cdot (x \overleftarrow{\nu}_t) \\ \overleftarrow{\nu}_0 = \gamma \end{cases}$$



Main Result

Theorem (D., Motsch, Parulekar, Porteous, Shakkottai)

$$\mu_y(x) = \omega(x) \nu_y^{DPS}(x). \quad (1)$$

*The weight ω admits a Feynman–Kac representation along the **backward path**:*

$$\omega(x) = \mathbb{E} \left[\frac{\vec{\mu}_T(Y_0)}{\gamma(Y_0)} \exp \left(- \int_0^T c_{DPS}(T-s, Y_s) ds \right) \mid Y_T = x \right]. \quad (2)$$



Main Result (continued)

Theorem (continued)

Equivalently, along the forward path:

$$\frac{1}{\omega(x)} = \mathbb{E}_{X \sim OU} \left[\frac{\gamma(X_T)}{\vec{\mu}_T(X_T)} \exp\left(\int_0^T c_{DPS}(s, X_s) ds\right) \mid X_0 = x \right]. \quad (3)$$



Main Result (continued)

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Consequence

Importance-weighting DPS samples by ω recovers μ_y exactly.



Discarding trajectories

Trajectories that have a small weight

$$\exp\left(-\int_0^T c_{DPS}(T-s, Y_s) ds\right)$$

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The catch is that posterior sampling with access to an oracle is exponentially bad in dimension for the general case. So any result that has a polynomial dependence on dimension needs to be for a specific class of simple priors ρ_* .

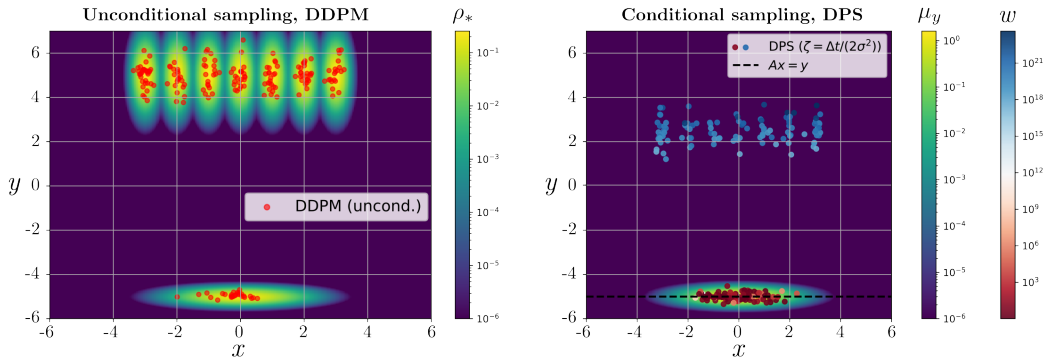


Figure: Left: prior and samples from the denoising SDE. Right: posterior for $R_y(x) = \|x_2 + 5\|^2$ and DPS samples; meaningless samples carry weights $w \sim 10^{20}$.



Numerical Stability

Algorithm 1 DPS

Require: $N, y, \{\zeta_i\}_{i=1}^N, \{\tilde{\sigma}_i\}_{i=1}^N$

1: $x_N \sim \mathcal{N}(0, I)$

2: **for** $i = N - 1$ **to** 0 **do**

3: $\hat{s} \leftarrow s_\theta(x_i, i)$

4: $\hat{x}_0 \leftarrow \frac{1}{\sqrt{\bar{\alpha}_i}}(x_i + (1 - \bar{\alpha}_i)\hat{s})$

5: $z \sim \mathcal{N}(0, I)$

6: $x'_{i-1} \leftarrow \frac{\sqrt{\bar{\alpha}_i(1-\bar{\alpha}_{i-1})}}{1-\bar{\alpha}_i}x_i + \frac{\sqrt{\bar{\alpha}_{i-1}}\beta_i}{1-\bar{\alpha}_i}\hat{x}_0 + \tilde{\sigma}_iz$

7: $x_{i-1} \leftarrow x'_{i-1} - \zeta_i \nabla_{x_i} \|y - \mathcal{A}(\hat{x}_0)\|^2$

8: **end for**

9: **return** x_0

▷ Initialize with Gaussian noise

▷ Evaluate the score oracle

▷ Compute $\mathbb{E}[X_0|X_t = x]$

▷ Integrate the score term

▷ Integrate the bias term **explicitly**



Numerical Stability

Algorithm 2 DPS

Require: $N, y, \{\zeta_i\}_{i=1}^N, \{\tilde{\sigma}_i\}_{i=1}^N$

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$$\zeta_i = \frac{1}{\|y - \mathcal{A}(\hat{x}_0)\|} \neq \Delta t$$



What is Δt ?

The time discretization is not fixed in time

$$\Delta t_i = -\frac{1}{2} \ln(1 - \beta_i) \quad (4)$$

The linear noise schedule of DDPM is given by

$$\beta_i = \beta_{min} + i \frac{\beta_{max} - \beta_{min}}{N} \approx 10^{-4} + 2i 10^{-5} \quad \text{for } i = 1, \dots, 1000,$$



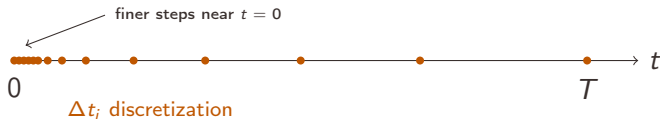
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Hidden annealing schedule

The actual surrogate path of DPS has the form

$$\mu_t^{DPS} = \frac{e^{-\eta_t R_y(\mathbb{E}[X_0|X_t=x])}}{Z} \rho_t$$

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The numerical integration of the biasing vector field does not satisfy the Forward Euler stability criterion!

Oscillations!

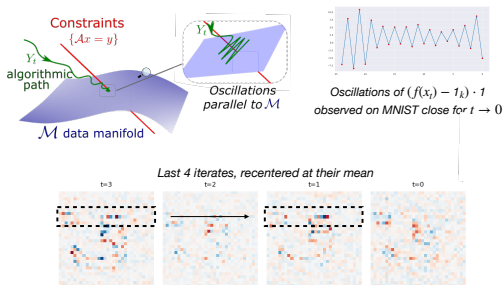


Figure: **(Top Left)** A pictorial depiction of instability. **(Top Right)** An exhibition of these oscillation on a posterior sampling task with an MNIST prior. **(Bottom)** A plot of the last four iterates of DPS, re-centered about their mean. The guidance tilted the distribution towards the digit 3. We observe periodic oscillations in pixel space.



Fixes



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- ▶ If the goal is not to enforce the hard constraint, practitioners (e.g. Nano Banana) turn off guidance in the last few steps.
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The analysis explains post-hoc the advantage of these choices made for deployed versions of the algorithms.



Outlook



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- ▶ The duality between reaction terms and drift has not been understood/exploited in the ML community. Possible applications to RL.
- ▶ The paradigm of sampling a prior might be changing from diffusion to gradient flows. Their posterior sampling analogue is not yet clear.



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Thank You!